VIII. Commonly Used Distributions (GSE 518)

Carlos Flores
Department of Economics
California Polytechnic State University*

Idea/Goal: Look at some commonly used distributions.

1. Special Distributions: Introduction

The special distributions we look at are *families of distributions*: their pdf/pmf is a function of one or more parameters.

- If we change the parameter(s) $\Rightarrow$ we change the location and/or shape of the pmf/pdf.

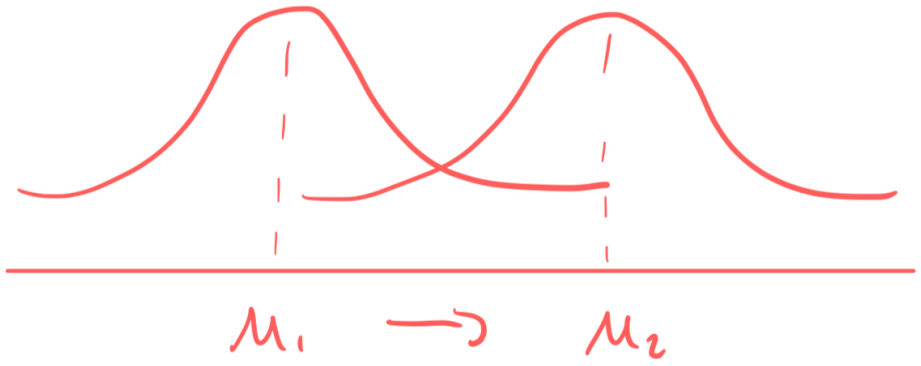
Example: $N(\mu, \sigma^2)$. For a Normal or Gaussian distribution, μ determines the *location* and σ^2 the *shape* of the distribution.

See figure below (*Note:* There is a typo in the figure, the more concentrated distribution should read "lower variance", and the more spread one should read "higher variance".)

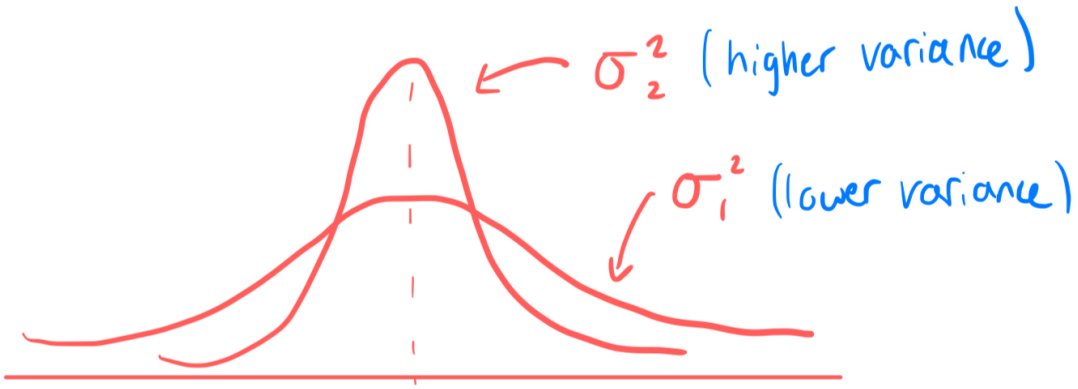
*MSQE Alumni Hunter Happersett contributed significantly to writing these lecture notes in Latex.

$N(\mu, \sigma^2)$

if we change μ :



if we change σ^2



2. Discrete Distributions

2.1. Discrete Uniform Distribution

A random variable X has a discrete uniform distribution if for a sample space $S_X = \{x_1, x_2, ..., x_N\}$, we have:

$$f_X(x) = \begin{cases} Pr(X = x_i) = \frac{1}{N} & \text{if } x_i \in S_X \\ 0 & \text{otherwise} \end{cases}$$

Notes:

- It assigns equal mass (or probability) to each point.
- If $S_X = \{1, 2, ..., N\}$, then $E(X) =$? and $Var(X) =$? (see Problem Set 4)
- Example: Toss a die $\implies$ $S_X = \{1, 2, 3, 4, 5, 6\}$.
- Mean and Variance:

2.2. Bernoulli Distribution

A r.v. X follows a Bernoulli distribution, or $X \sim Bernoulli(p)$, if for $p \in [0, 1]$

$$f_X(x) = \begin{cases} p^x(1-p)^{1-x} & \text{if } x \in \{0, 1\} \\ 0 & \text{o/w} \end{cases}$$

Notes:

- Note that $Pr(X = 1) = p \quad \text{and} \quad P(X = 0) = 1 - p$
- Usually we think of $X = 1$ as "success" and $X = 0$ as "failure".
- $E(X) = p \quad \text{and} \quad Var(X) = p \cdot (1 - p)$

- Example: Toss a coin and define X as:

$$X = \begin{cases} 1 & \text{if heads} \\ 0 & \text{if tails} \end{cases}$$

In this case, $p = Pr(X = 1) = \frac{1}{2}$.

- The Bernoulli distribution is useful to model *binary* (also called *dummy* or *indicator*) variables, i.e., variables that can take the values of 0 or 1. Examples:

- Whether an individual is employed or not
- Whether a team will win a game or not
- Whether a person will default on a loan or not

Note: Logit and Probit models are based on this distribution (*later in econometrics*)

Key General Note: *Keep in mind that $E(X)$ and $Var(X)$ are always functions of the parameters of the distribution.*

2.3. Binomial Distribution

A random variable $X \sim Binomial(n, p)$ for $n \in \mathbb{N}$ and $p \in [0, 1]$ if:

$$f_X(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & \text{if } x \in \{0, 1, \ldots, n\} \\ 0 & \text{o/w} \end{cases}$$

where $\binom{n}{x} = \frac{n!}{(n-x)!x!}$ and "!" is the factorial operator defined as $m! = m \cdot (m-1) \cdot (m-2) \cdots 1$.

Notes:

-The support is $S_X = \{0, 1, \ldots, n\}$.

- We can think of the Binomial distribution as n independent Bernoulli trials with constant p. Then, $X =$ number of ones, i.e.,

$$X = \sum_{i=1}^{n} Y_i \quad \text{with } Y_i \sim Bernoulli(p)$$

- Example: Flip a coin 100 times and let $X =$ Number of Heads. Here $n = 100$ and $p = 1/2$.

- $Binomial(1, p) = Bernoulli(p)$

- Mean and Variance:

$$E(X) = np, \quad \text{Var}(X) = np(1-p)$$

- Examples of possible applications:

- In a basketball season with 82 games, let X equal the number of games won by a team ($n = 82$ and p equal the probability of winning a game).
- Given a flight with 100 tickets sold, let X equal the number of people that shows up to take the flight.

- *Intuition* about the pmf formula: $\binom{n}{x} \cdot p^x \cdot (1-p)^{n-x}$

$$p^x(1-p)^{n-x} = \quad \text{probability of an outcome with } x \text{ ones and } (n-x) \text{ zeroes}$$

$$\binom{n}{x} = \frac{n!}{(n-x)!x!} = \quad \text{number of outcomes with } x \text{ ones and } (n-x) \text{ zeroes}$$

(Hence, the second term is only "counting" outcomes to ensure we "cover" all cases)

(See example from lecture)

- *General Note* (Aside): Be careful when getting probabilities in the discrete case—equality sign does matter! Example: Assume the sample space is $S_X = \{0, 1, 2, ...\}$, then, we can get $Pr(X = 2)$ as:

$$\mathbb{P}(X \geq 2) = 1 - \mathbb{P}(X < 2) = 1 - \mathbb{P}(X \leq 1)$$

2.4. Geometric Distribution

A random variable $X \sim$ Geometric(p) for $p \in [0, 1]$ if:

$$f_X(x) = P(X = x) = \begin{cases} p(1-p)^{x-1} & \text{if } x = 1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$$

Notes:

- Think of it as n independent Bernoulli(p) trials with constant p. Then $X = \#$ of trials until the first one (or "success") is achieved.
- Example: $X = \#$ of flips of a coin needed to get the first heads (one).
- Used to model "lifetime or time until failure or success". Examples:

  - # of weeks until a person finds employment.
  - # of games until the first win.

- Mean and Variance:

$$E(X) = \frac{1}{p}, \quad \text{Var}(X) = \frac{1-p}{p^2}$$

2.5. Negative Binomial Distribution

A random variable $X \sim NB(r, p)$, for positive integer r and $p \in [0, 1]$ if

$$f_X(x) = \begin{cases} \binom{x-1}{r-1} \cdot p^r \cdot (1-p)^{x-r} & \text{for } x = r, r+1, r+2, \ldots \\ 0 & \text{otherwise} \end{cases}$$

where $\binom{a}{b} = \frac{a!}{(a-b)!b!}$.

Notes:

- Think of this as n independent Bernoulli(p) trials with p constant. $X = \#$ of trials for the r-th "one". Examples:

  - # of flips until we get the r-th heads.
  - # of games until the r-th win.

- Note that: NB$(1, p)$ = Geometric(p)
- Mean and Variance: $E(X) = \frac{r(1-p)}{p}$ and $Var(X) = \frac{r(1-p)}{p^2}$

2.6. Hypergeometric Distribution

A random variable $X \sim$ Hypergeometric(N, M, K) if $1 \leq K \leq N$, $0 \leq M \leq N$.

$$f_X(x) = \begin{cases} \frac{\binom{M}{x}\binom{N-M}{K-x}}{\binom{N}{K}} & \text{for } x = 0, 1, \ldots, K \\ 0 & \text{otherwise} \end{cases}$$

Notes:

- Think of it as sampling K times without replacement (so probabilities change) from a box with N balls, M of which are red and $N - M$ are blue. Then, $X = \#$ of red balls in the sample.

- This distribution is useful when sampling from a *finite* population. Examples:

- Hire K people from a pool of N applicants. M are a good fit and $N - M$ are a bad fit. $X = \#$ of hires that are a good fit.

- Used a lot for quality control. Consider a lot of N items (e.g., bottles of wines or iphones) and checking quality control by choosing at random K of them. Let M be the # of defective products in the lot. Then, $X = \#$ of defective items in our sample of K items.

2.7. Poisson Distribution

A random variable $X \sim \text{Poisson}(\lambda)$ for $\lambda > 0$ if

$$f_X(x) = \begin{cases} \frac{e^{-\lambda}\lambda^x}{x!} & \text{for } x = 0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

Notes:

- It has many uses:

- To model *"time until success (or failure)"*, e.g., # of months unemployed until finding a job; # of loses until a team gets its first win.

- To model *"count data" (Poisson Regression)*, e.g., # of kids in a family; # of car accidents in a day.

- To model *spatial distributions*, e.g., distribution of stores in a city, distribution of fish in a lake.

- It gives a good approximation to a $Binomial(n, p)$ for large n and small p, where $\lambda = n \cdot p$.

- Mean and Variance: $E(X) = \lambda \quad \text{and} \quad Var(X) = \lambda$.

- Note: The fact that the mean equals the variance can be a disadvantage for modeling.

3. Continuous Distributions

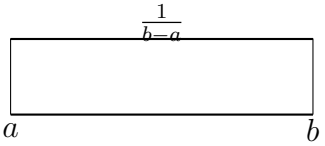
3.1. Uniform Distribution

A random variable $X \sim \text{Uniform}(a, b)$ if:

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Notes:

- Graph of pdf of Uniform(a, b):



- Uniform$(0, 1)$ is used a lot to generate random samples (later: for simulations, bootstrap).

- $E(X)$, Var(X): see P.S. #4.

3.2. Gamma Distribution

A random variable $X \sim \text{Gamma}(\alpha, \beta)$ for $\alpha, \beta > 0$ if:

$$f_X(x) = \begin{cases} \frac{x^{\alpha-1} e^{-\frac{x}{\beta}}}{\Gamma(\alpha)\beta^{\alpha}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Notes:

- $\Gamma(\alpha)$, the *gamma function* is defined as:

$$\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u}\, du$$

- Some *Properties of the gamma function*:

(i) $\Gamma(1) = 1$

(ii) $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

(iii) $\Gamma(\alpha + 1) = \alpha\Gamma(\alpha)$ for $\alpha > 0$

(iv) $\Gamma(\alpha) = (\alpha - 1)!$ for positive integer α

- Some *Intuition on the gamma function*:

- Can think of the gamma function as an extension of factorials to the continuous case.
- Informally, an analogous to the discrete case, it ensures we cover (or "count") all possible cases.
- It is a function that ensures that the pdf integrates to 1: $\int_{x \in X} f_X(x)\, dx = 1$

- *Special Cases* of the gamma distribution:

- Gamma$(1, \beta)$ = Exponential(β)
- Gamma$\left(\frac{p}{2}, 2\right)$ = Chi-square(p) for positive integer p.

- It may take many shapes, which makes it good for modeling.
- $E(X) = \alpha\beta$, $\text{Var}(X) = \alpha\beta^2$.

3.3. Chi-square Distribution

A random variable $X \sim \chi^2_{(p)}$ for a positive integer p (called "degrees of freedom") if

$$f_X(x) = \frac{x^{\frac{p}{2}-1}e^{-\frac{x}{2}}}{\Gamma(\frac{p}{2})2^{\frac{p}{2}}} \quad \text{if } x > 0$$

Notes:

- A $\chi^2_{(p)}$ equals a Gamma$\left(\frac{p}{2}, 2\right)$.

- Used a lot for hypothesis testing (later).

- Key Property: If $X_1, X_2, \ldots, X_n$ are independent random variables each with distribution $X_i \sim \chi^2_{(p_i)}$, then:

$$\sum_{i=1}^{n} X_i \sim \chi^2_{\left(\sum_{i=1}^{n} p_i\right)}$$

In words, the sum of independent Chi-squared random variables follows a Chi-squared distribution (and we sum the degrees of freedom).

For example, if $X_1 \sim \chi^2_1$, $X_2 \sim \chi^2_2$, and X_1 and X_2 are independent $\Rightarrow X_1 + X_2 \sim \chi^2_{(1+2)}$.

- $E(X) = p$ and $Var(X) = 2p$.

3.4. Exponential Distribution

A random variable $X \sim$ Exponential(λ) for $\lambda > 0$ if:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Notes:

-Used to model time until an event occurs (e.g., time until finding a job, time until a light bulb burns out).

- Common distribution used in *Duration* or *Hazard* Models.
- In that sense, analogous to Poisson distribution, but now when working with time in continuous time.

- $E(X) = \lambda$ and $Var(X) = \lambda^2$.

3.5. Beta Distribution

A random variable $X \sim \text{Beta}(\alpha, \beta)$ if:

$$f_X(x) = \begin{cases} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1} & \text{if } 0 < x < 1 \text{ and } \alpha, \beta > 0 \\ 0 & \text{otherwise} \end{cases}$$

Notes:

- Good distribution to model variables between 0 and 1 (e.g., proportions).

- The Beta distribution can take many shapes (nice for modeling).

- A disadvantage for modeling is that X cannot take a value of zero.

- $\text{Beta}(1, 1) = \text{Uniform}(0, 1)$

- If $X_1 \sim Gamma(\alpha, 1)$ and $X_2 \sim Gamma(\beta, 1)$, then $\frac{X_1}{X_1+X_2} \sim Beta(\alpha, \beta)$

- $E(X) = \frac{\alpha}{\alpha+\beta}$ and $Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.

3.6. Normal Distribution

A random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ if:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{for } x \in \mathbb{R}, \sigma > 0$$

Notes:

- Key distribution in statistics:

  - Nice/easy distribution to work with (e.g., no gamma function!).
  - Nice bell shape; good for modeling (sometimes).
  - Used a lot for hypothesis testing (later).
  - *Central Limit Theorem:* Used to approximate the distribution of many (if not most) estimators in large samples (later).

- $E(X) = \mu, \quad \text{Var}(X) = \sigma^2$

- *Very Useful Result (see Assignment 3 for proof):* If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = aX + b$, then $Y \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$

  - In words, a linear function of a normally distributed random variable also follows a normal distribution.
  - Why does it have that mean and variance? See the results from the Expectations notes.
  - *Example:* Let $X \sim \mathcal{N}(1, 4)$ and $Y = 2X + 3$. What is the distribution of Y?

$$E(Y) = E(2X + 3) = 2E(X) + 3 = 2(1) + 3 = 5$$

$$\text{Var}(Y) = \text{Var}(2X + 3) = 4\text{Var}(X) = 4(4) = 16$$

$$Y \sim \mathcal{N}(5, 16)$$

- *Very Important Special case (used to standardize a variable):* Let $Z = \frac{X-\mu}{\sigma}$, then $Z \sim \mathcal{N}(0, 1)$. This distribution is called the *Standard Normal distribution*.

- *Some Properties of the Standard Normal distribution.* Usually we write $\phi(z)$ for the pdf of $\mathcal{N}(0, 1)$ and $\Phi(z)$ for the cdf of $\mathcal{N}(0, 1)$. Properties of $\mathcal{N}(0, 1)$:

1. $\phi(z) = \phi(-z)$ (true for any symmetric distribution)
2. $\Phi(-z) = 1 - \Phi(z)$ (true for any symmetric distribution)
3. If $Z \sim \mathcal{N}(0, 1)$, then $Z^2 \sim \chi^2_1$ (see P.S. 5).

- There is no closed form formula for the cdf of a $\mathcal{N}(\mu, \sigma^2)$

- Then, how do we calculate probabilities? Usually (or "before google") we standardize the random variable and look at a table of probabilities for a $\mathcal{N}(0, 1)$ distribution.
- Example: Let $X \sim \mathcal{N}(\mu, \sigma^2)$, find $P(X \leq a)$.

$$P\left(Z \leq \frac{a - \mu}{\sigma}\right)$$

3.7. t-distribution

A random variable $X \sim t$-distribution with $r \in \mathbb{N}$ degrees of freedom if:

$$f_X(x) = \frac{\Gamma\left(\frac{r+1}{2}\right)}{\sqrt{r\pi}\,\Gamma\left(\frac{r}{2}\right)} \cdot \left[1 + \frac{x^2}{r}\right]^{-\frac{r+1}{2}} \quad \text{for } x \in \mathbb{R}$$

Notes:

- Looks like a normal distribution graphically but with thicker tails.

- $E(X) = 0$ if $r > 1$ and $\text{Var}(X) = \frac{r}{r-2}$ if $r > 2$. mgf does not exist.

- A t distribution with $r = 1$ is called a Cauchy distribution. $E(X)$ does not exist for this distribution.

- Used a lot for hypothesis testing (later).

- As $r \to \infty$, it approaches $N(0, 1)$.

3.8. F-distribution

$X \sim F(r_1, r_2)$, r_1, r_2 positive integers and called "degrees of freedom" if:

$$f_X(x) = \begin{cases} \text{See book} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Notes:

- r_1 and r_2 are called *degress of freedom*

- $E(X)$ exists if $r_2 > 2$ and $\text{Var}(X)$ exists if $r_2 > 4$

- Used a lot for hypothesis testing (later).

- If you have $F(1, r_2) = t^2(r_2)$ where t is a t distribution with r_2 degrees of freedom.

3.9. Laplace or Double Exponential Distribution

$$f(x) = \frac{1}{2\sigma} e^{-\frac{|x-\mu|}{\sigma}} \quad \text{for } x \in \mathbb{R}$$

Notes:

- It looks like a normal distribution graphically.

- Useful for "median regression" and its extension to "quantile regression" (later).

3.10. Log Normal Distribution

A random variable X is log normal if:

$$\log(X) \sim N(\mu, \sigma^2) \quad \text{for } x > 0$$

Notes:

- Useful if you have outcomes skewed to the right and $x \geq 0$ (e.g., income).